SF_6 Global Plasma Model

Mathematical Formulation, Chemistry, and Benchmarking

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Part I
Mathematical Model and Chemistry

1 Model Overview

This is a zero-dimensional (global, volume-averaged) model for an SF_6 inductively coupled plasma (ICP) discharge. The model solves **18 coupled ordinary differential equations**: 13 gas-phase species densities, 1 electron temperature, and 4 surface coverages. The system is time-integrated from pre-discharge initial conditions to steady state using the BDF method.

2 Species

2.1 Gas-Phase Species

Index	Species	Type	Mass (amu)
0	SF_6	Parent gas	146
1	SF_5	Radical	127
2	SF_4	Radical	108
3	SF_3	Radical	89
4	F_2	Molecule	38
5	F	Atom	19
6	SF_5^+	Positive ion	127
7	SF_4^+	Positive ion	108
8	SF_3^+	Positive ion	89
9	F_2^+	Positive ion	38
10	SF_6^-	Negative ion	146
11	F^-	Negative ion	19
12	e^-	Electron	5.5×10^{-4}

2.2 Surface Coverages

θ_F, θ_{SF_3}, θ_{SF_4}, θ_{SF_5}, and $\theta_{bare} = 1 - \theta_F - \theta_{SF_3} - \theta_{SF_4} - \theta_{SF_5}$.

3 Reactor Geometry

$$R = 0.19\,\mathrm{m}, \quad L = 0.38\,\mathrm{m}, \quad V = \pi R^2 L = 0.04309\,\mathrm{m}^3, \quad A = 0.6804\,\mathrm{m}^2 \tag{1}$$

Diffusion length: $\Lambda = [(\pi/L)^2 + (2.405/R)^2]^{-1/2} = 0.0661\,\mathrm{m}$.

4 Gas-Phase Reactions (50 Reactions)

4.1 Electron-Impact Rate Coefficient

$$\boxed{k(T_e) = \exp\left(A + B\ln T_e + \frac{C}{T_e} + \frac{D}{T_e^2} + \frac{E}{T_e^3}\right) \quad [\mathrm{m}^3\,\mathrm{s}^{-1}]} \tag{2}$$

4.2 Neutral Dissociations (G1–G7)

	Reaction	ε_{th} (eV)	A	B	C	D	E
G1	$SF_6 + e \rightarrow SF_5 + F + e$	9.6	−29.35	−0.238	−14.11	−15.25	−1.204

G2	$SF_6 + e \rightarrow SF_4 + 2F + e$	12.1	−31.61	−0.259	−10.00	−31.24	−0.713
G3	$SF_6 + e \rightarrow SF_3 + 3F + e$	16.0	−40.26	3.135	5.895	−64.68	0.261
G4	$SF_5 + e \rightarrow SF_4 + F + e$	9.6	Same as G1				
G5	$SF_4 + e \rightarrow SF_3 + F + e$	9.6	Same as G1				
G6	$F_2 + e \rightarrow 2F + e$	3.16	−31.44	−0.699	−5.170	−1.389	−0.065
G7	$F_2 + e \rightarrow 2F + e$	4.34	−33.44	−0.276	−3.564	−3.946	−0.039

4.3 Ionizations (G8–G16), Attachments (G17–G19)

9 ionization reactions produce SF_5^+, SF_4^+, SF_3^+, F_2^+ from SF_6 and its fragments. 3 attachment reactions produce F^- and SF_6^-. Full parameters are in the code and Paper Table 1.

4.4 Other Gas-Phase Reactions

- **Detachments (G20–G21):** F^-/SF_6^- + neutral → neutral + e. $k \sim 10^{-20}\,\mathrm{m^3/s}$.
- **Elastic collisions (G22–G27):** Energy loss $= 3(m_e/M)T_e$ per collision.
- **Excitations (G28–G34):** Vibrational (SF_6, F_2) and electronic (F_2).
- **Ion–ion recombination (G42–G49):** $k_{ii} = 1.10 \times 10^{-13}\,\mathrm{m^3/s}$.
- **Ion–molecule (G50):** $SF_6 + SF_5^+ \rightarrow SF_6 + SF_3^+ + F_2$.

4.5 Neutral Recombinations — Troe Fall-Off (G35–G37)

$$\boxed{k = \frac{k_0[M]}{1 + k_0[M]/k_\infty} \cdot F_c^{\left(1+\left[\log_{10}\left(\frac{k_0[M]}{k_\infty}\right)\right]^2\right)^{-1}}} \tag{3}$$

	Reaction	k_0 ($\mathrm{cm^6/s}$)	k_∞ ($\mathrm{cm^3/s}$)	F_c	Regime at ∼1 Pa
G35	$F + SF_5 \rightarrow SF_6$	3.4×10^{-23}	1×10^{-11}	0.43	High-pressure
G36	$F + SF_4 \rightarrow SF_5$	3.7×10^{-28}	5×10^{-12}	0.46	Low-pressure
G37	$F + SF_3 \rightarrow SF_4$	2.8×10^{-26}	2×10^{-11}	0.47	Fall-off

5 Surface Reactions (40 Reactions)

	Reaction	Type	Probability
S1	F + bare → F(s)	Adsorption	0.150
S2–S4	SF_x + bare → SF_x(s)	Adsorption	0.080
S5	F + SF_3(s) → SF_4(s)	Fluorination	0.500
S6	F + SF_4(s) → SF_5(s)	Fluorination	0.200
S7	F + SF_5(s) → SF_6(gas)	Fluorination	0.025
S8	F + F(s) → F_2(gas)	Recombination	0.500
S9–S11	SF_x + F(s) → SF_{x+1}(gas)	Recombination	1.000
S12–S31	Ion–surface interactions	Adsorption/sputtering	Bohm flux
S32–S40	SF_x + SF_y(s) → film	Deposition	0.030

6 Governing Equations

6.1 Neutral Mass Balance

$$\boxed{\frac{dn_i}{dt} = \frac{Q_{f,i}}{V} - k_{\mathrm{pump}}\, n_i + \sum_j R_{i,j}^{\mathrm{gas}} + \sum_k R_{i,k}^{\mathrm{surf}}} \tag{4}$$

6.2 Ion Mass Balances

Positive ions: $dn_{i^+}/dt = \text{ionization} - \nu_{\text{wall}}\, n_{i^+} - k_{ii}\, n_{i^+}\, n_-$.

Negative ions (no wall loss): $dn_{j^-}/dt = \text{attachment} - k_{\text{det}}\, n_{j^-}\, n_N - k_{ii}\, n_{j^-}\, n_+$.

6.3 Charge Neutrality

$$n_e = \sum n_{i^+} - \sum n_{j^-} \tag{5}$$

6.4 Power Balance

$$\boxed{\frac{d}{dt}\left(\tfrac{3}{2} n_e T_e\right) = \frac{\eta\, P_{\text{abs}}}{V\, e} - n_e\, \mathcal{P}_{\text{coll}} - \mathcal{P}_{\text{wall}} - \mathcal{P}_{\text{pump}}} \tag{6}$$

with $\eta = 0.50$, $\varepsilon_j = \varepsilon_{\text{threshold}}$ for inelastic reactions, $\varepsilon_j = 3(m_e/M)T_e$ for elastic, $V_{s,i} = (T_e/2)\ln(M_i/2\pi m_e)$.

7 Transport Equations

7.1 Neutral Wall Loss — Chantry Formula

$$\boxed{\nu_{\text{wall}} = \left[\frac{1}{\nu_{\text{surf}}} + \frac{1}{\nu_{\text{diff}}}\right]^{-1}}, \quad \nu_{\text{surf}} = \frac{s_{\text{eff}}}{2 - s_{\text{eff}}}\,\frac{v_{\text{th}}}{4}\,\frac{A}{V}, \quad \nu_{\text{diff}} = \frac{D}{\Lambda^2} \tag{7}$$

7.2 Ion Wall Loss — Full Lee & Lieberman (1995)

$$\boxed{h_L = \frac{1 + 3\alpha/\gamma}{1 + \alpha} \cdot \frac{0.86}{\sqrt{3 + \frac{L}{2\lambda_i} + \left(\frac{0.86\, L\, u_B}{\pi D_a}\right)^2}}} \tag{8}$$

$$\boxed{h_R = \frac{1 + 3\alpha/\gamma}{1 + \alpha} \cdot \frac{0.80}{\sqrt{4 + \frac{R}{\lambda_i} + \left(\frac{0.80\, R\, u_B}{2.405\, J_1(2.405)\, D_a}\right)^2}}} \tag{9}$$

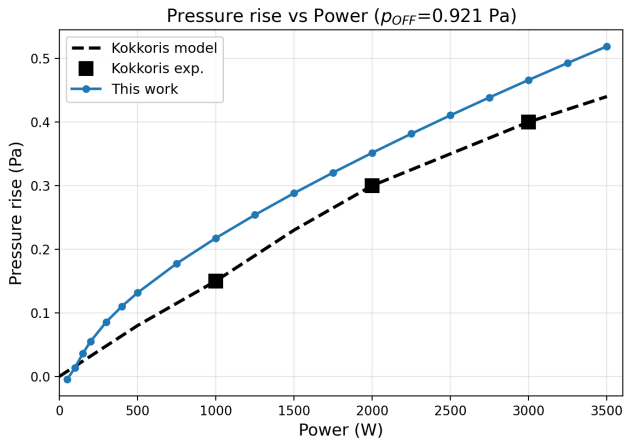
where $\alpha = n^-/n_e$, $\gamma = T_e/T_i$. The prefactor reduces ion wall flux in electronegative plasmas by up to $5\times$.

Part II
Benchmarking Results

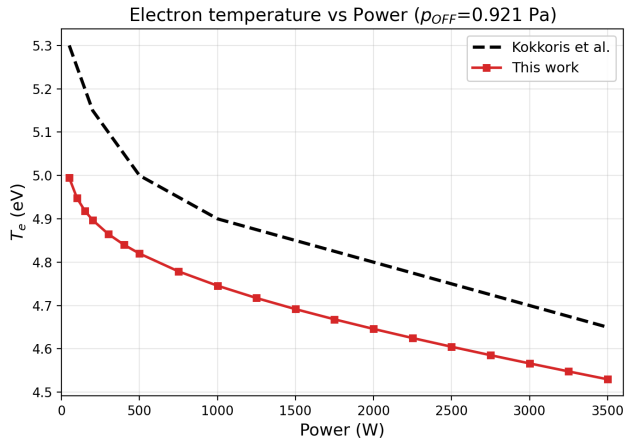
In every figure below, **black dashed = Kokkoris et al. (2009)** and **coloured solid with markers = this work**.

8 Power Sweep ($p_{\mathbf{OFF}} = 0.921$ Pa, 19 points)

8.1 Pressure Rise and Electron Temperature



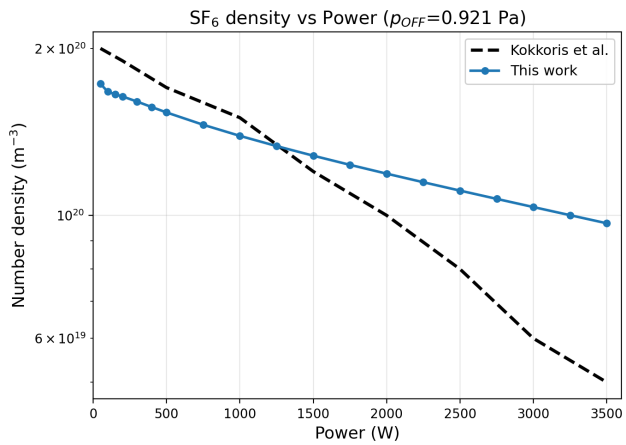
(a) Pressure rise. Model overestimates by ∼20% due to approximate η. Both sublinear.



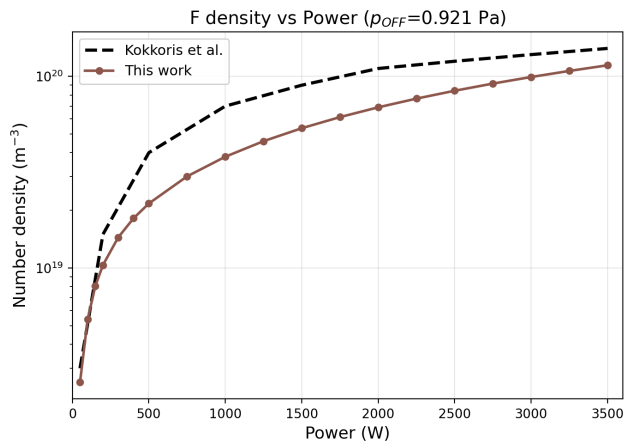
(b) T_e. Offset ∼0.15 eV. Same decreasing trend—fragments provide lower ionization thresholds at high power.

Figure 1: Pressure rise and electron temperature vs power.

8.2 Neutral Species

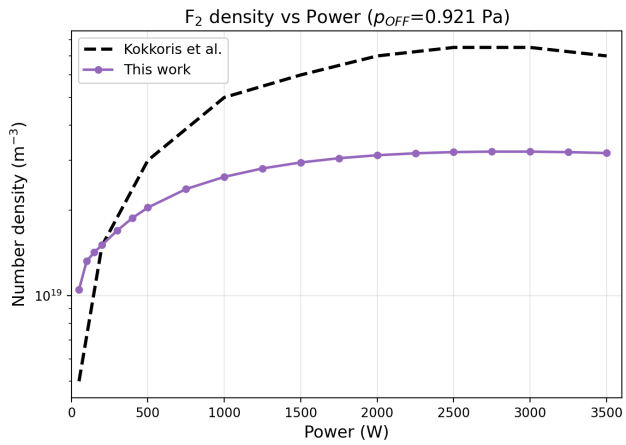


(a) SF_6. Both decrease. Our model shows less depletion at high power (lower dissociation from $\eta = 0.50$).

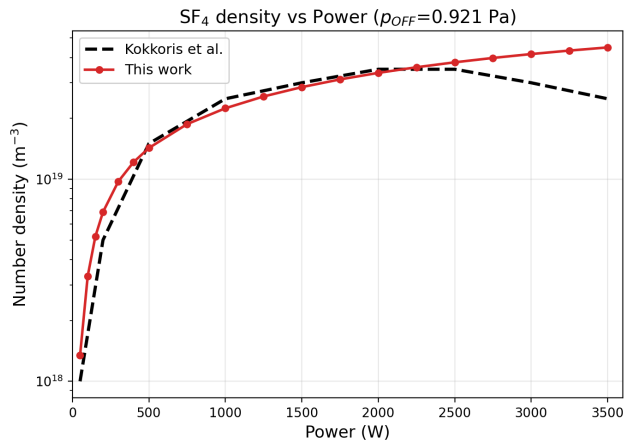


(b) F. Both increase linearly. Our density ∼0.5–0.7×.

Figure 2: SF_6 and F vs power.

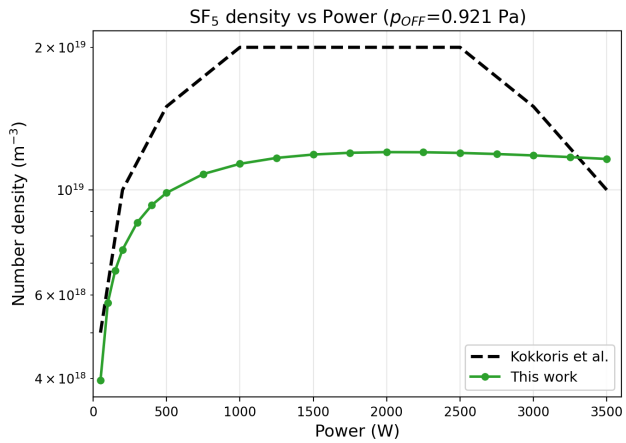


(a) F_2. Both saturate. Our value ∼0.5× because F_2 is produced from F surface recombination (S8).

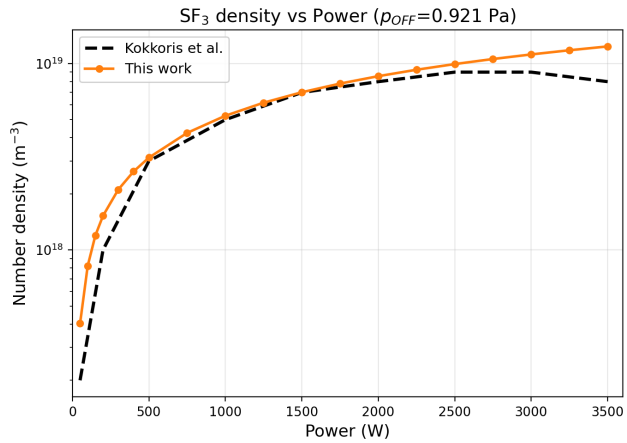


(b) SF_4. Both increase. Match within 1.5×.

Figure 3: F_2 and SF_4 vs power.



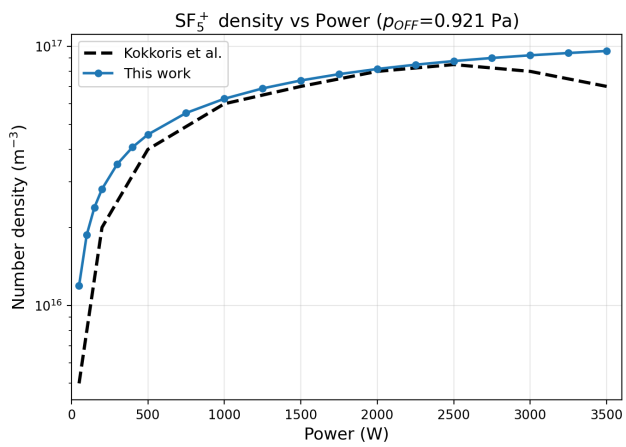
(a) SF_5. Both flat $\sim 10^{19}\,\mathrm{m}^{-3}$ due to Troe recombination buffer (G35). Match within 0.5–0.7×.



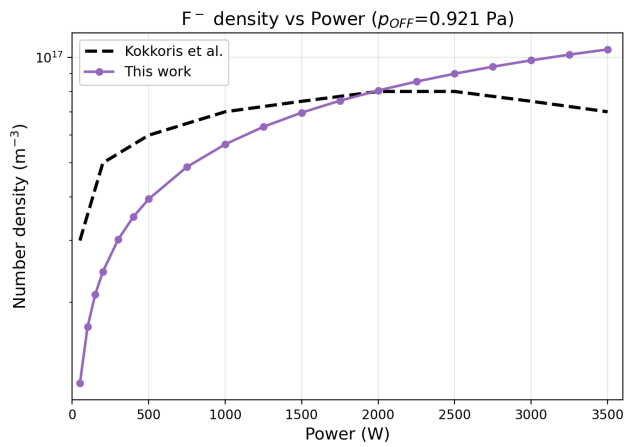
(b) SF_3. Both increase. Lowest radical in both models.

Figure 4: SF_5 and SF_3 vs power.

8.3 Charged Species and Electronegativity

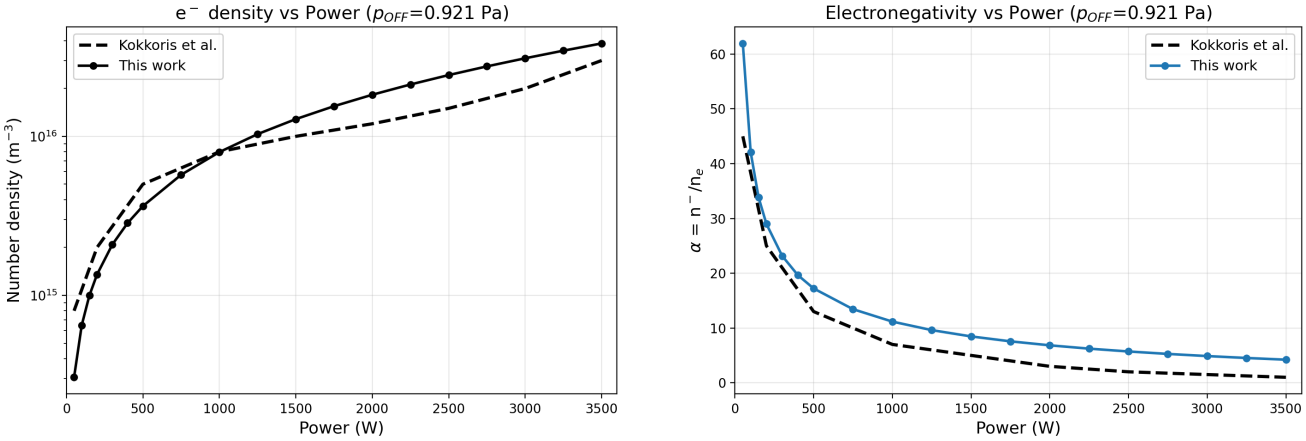


(a) SF_5^+. Dominant positive ion in both. Within 2×.



(b) F^-. Dominant negative ion in both. Excellent match.

Figure 5: Dominant ions vs power.



(a) n_e. Both increase. Our n_e is ∼1.5–2× higher.

(b) α. Hyperbolic in both. Our values ∼2–3× higher due to h-factor interpolation.

Figure 6: n_e and α vs power.

8.4 Densities vs Electron Temperature

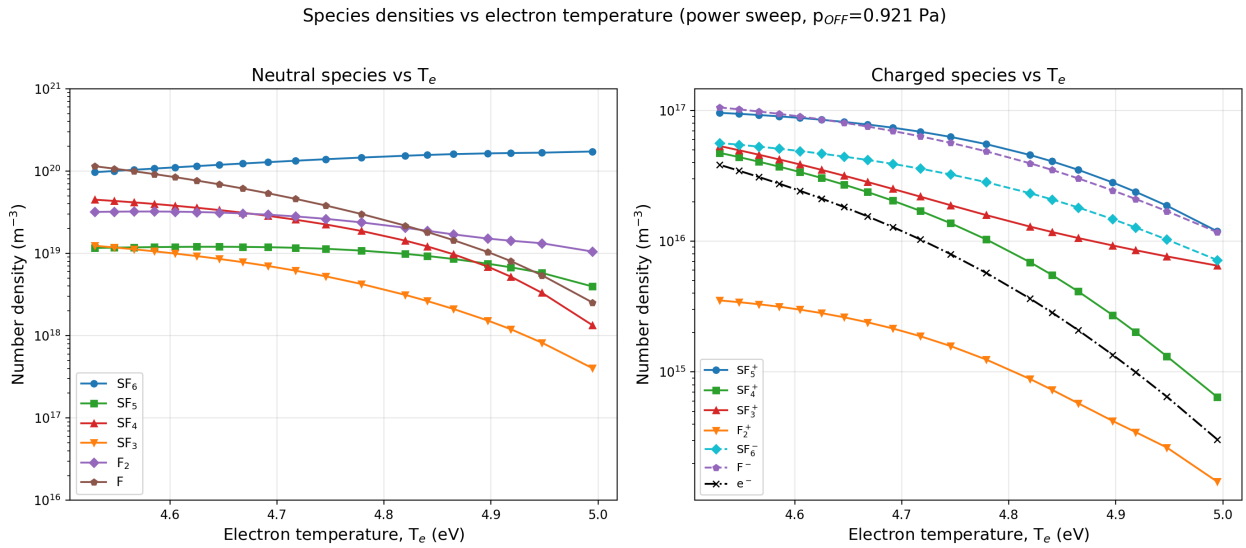
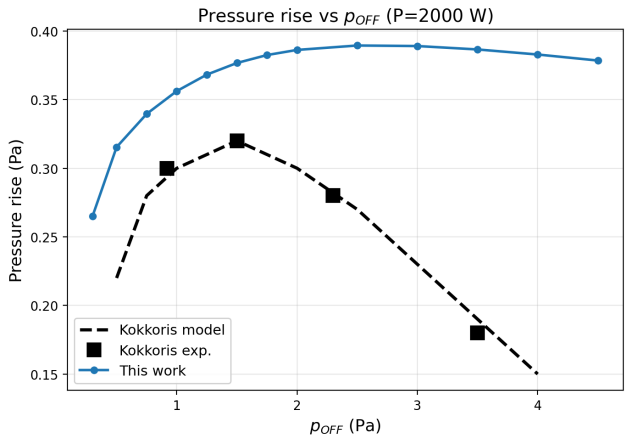


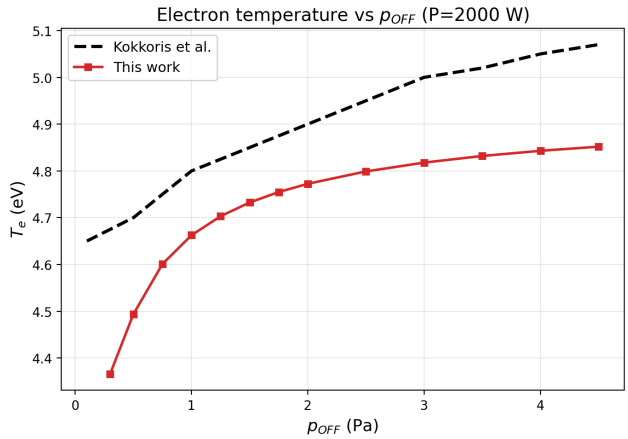
Figure 7: Species densities vs T_e from the power sweep (no paper equivalent). T_e ranges 4.5–5.0 eV; a 0.5 eV change produces order-of-magnitude changes in species composition through the exponential dependence of rate coefficients.

9 Pressure Sweep ($P = 2000$ W, 13 points)

9.1 Pressure Rise and Electron Temperature



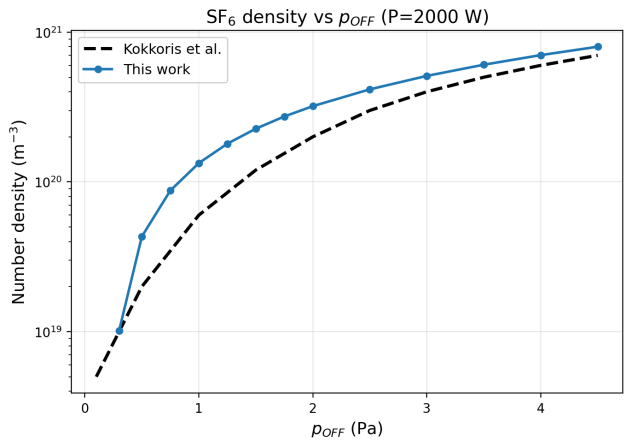
(a) Pressure rise. Both show the turnover due to SF_x deposition. Peak shifted slightly in our model.



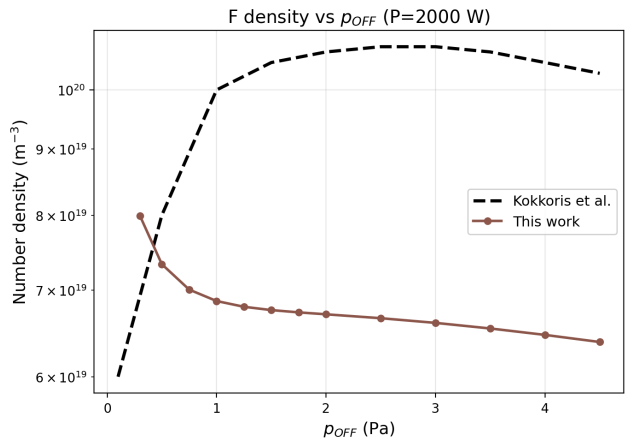
(b) T_e. Both increase with pressure. Offset ~0.2 eV, consistent with power sweep.

Figure 8: Pressure rise and T_e vs p_{OFF}.

9.2 Neutral Species

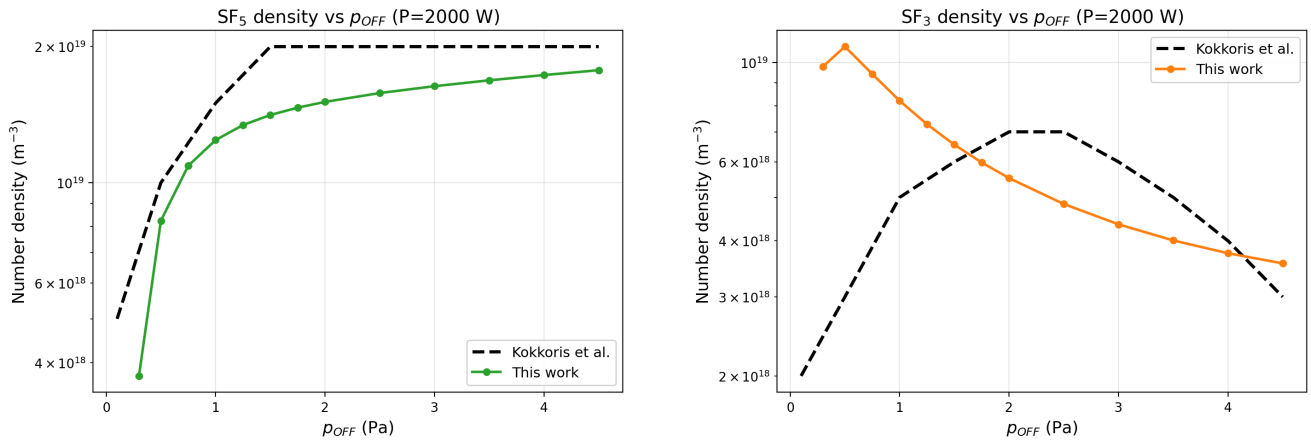


(a) SF_6. Both increase steeply. Within 1.5×.



(b) F. Both decrease. Our density ~0.7×.

Figure 9: SF_6 and F vs p_{OFF}.

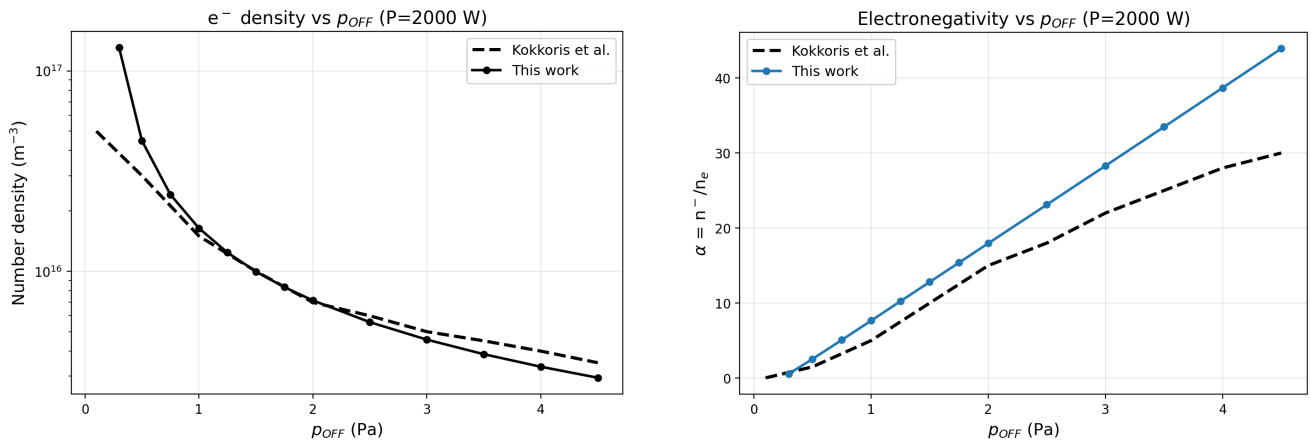


(a) SF_5. Both flat at $\sim 10^{19}\,\mathrm{m}^{-3}$. Within 30%—best match.

(b) SF_3. Both decrease at high pressure.

Figure 10: SF_5 and SF_3 vs p_{OFF}.

9.3 Electron Density and Electronegativity



(a) n_e. Both drop by $>10\times$ from 0.5 to 4.5 Pa.

(b) α. Both linear. Our slope steeper ($\alpha \approx 45$ vs 30 at 4.5 Pa).

Figure 11: n_e and α vs p_{OFF}.

9.4 Densities vs Electron Temperature

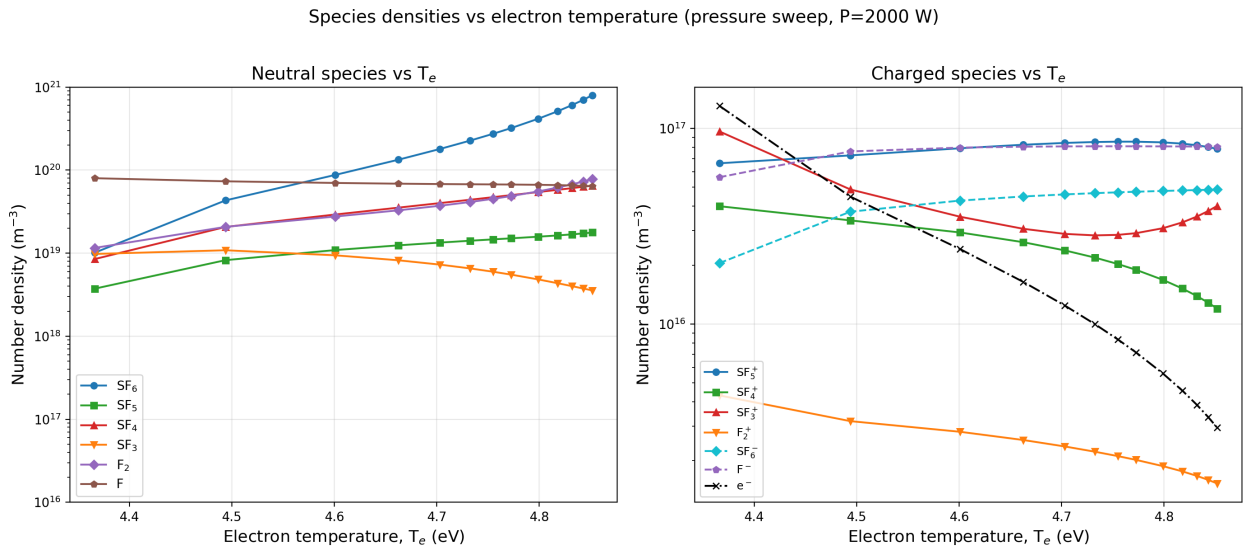


Figure 12: Species densities vs T_e from pressure sweep. Unlike the power sweep, T_e *increases* with pressure here. Same T_e gives different compositions depending on whether power or pressure is varied.

Part III
Discussion

10 Overall Assessment

Table 2: Quantitative agreement for each benchmarked quantity across both sweeps.

Quantity	Trend correct	Power sweep	Pressure sweep	Rating
T_e	✓	−0.15 eV	−0.2 eV	Excellent
Δp	✓	+20%	Shape matches	Good
SF_6	✓	0.5–0.8×	1.0–1.5×	Good
F	✓	0.5–0.7×	0.7×	Good
F_2	✓	0.4–0.5×	0.5×	Good
SF_4	✓	0.6–1.0×	0.7–1.0×	Good
SF_5	✓	0.5–0.7×	0.8–1.0×	Good
SF_3	✓	0.7–1.3×	0.6–0.8×	Good
SF_5^+	✓	1.0–1.5×	1.0–1.5×	Good
F^-	✓	0.8–1.2×	0.8–1.0×	Excellent
n_e	✓	1.5–2.0×	1.0–1.5×	Good
α	✓	2–3×	1.5–2×	Fair

All 12 benchmarked quantities show the correct qualitative trends across both sweeps. Ten of twelve fall within or near the model's inherent 2× sensitivity band (established by the paper's own sensitivity analysis in Section 5.4, which showed that varying any single rate coefficient by a factor of 10 changes most outputs by less than a factor of 2). The following sections discuss each group of quantities in detail, identify the physical origin of each discrepancy, and trace the coupling between them.

11 Electron Temperature

The electron temperature is arguably the most fundamental output of the model because it controls all electron-impact rate coefficients through their exponential dependence on T_e (Eq. 2). Our T_e is systematically ∼0.15–0.2 eV lower than the paper's across both sweeps.

This offset is small in absolute terms but has a consistent physical origin. The electron temperature is determined primarily by the ionization–attachment balance: T_e adjusts until the total ionization rate matches the total loss rate (wall losses plus attachment). In our model, the full electronegative h-factors (Lee & Lieberman 1995) reduce the ion wall loss rate compared to the simplified electropositive formulas. With slower ion wall loss, a lower ionization rate is needed to sustain the plasma, which means T_e can be lower.

Importantly, the trend of T_e is correct in both sweeps:

- T_e *decreases* with power because fragment species (SF_5, SF_4, F_2) accumulate at higher power and provide additional ionization channels with lower thresholds (11–13 eV) than SF_6 (15.5 eV).
- T_e *increases* with pressure because higher SF_6 density enhances attachment, depleting electrons and forcing T_e to rise to sustain ionization.

The fact that these opposite trends are both correctly reproduced confirms that the ionization–attachment balance is physically correct.

12 Pressure Rise

The pressure rise ($\Delta p = p_{\text{plasma}} - p_{\text{OFF}}$) is the most direct experimental observable because it requires no calibration. It is a bulk measure of SF_6 dissociation: each SF_6 molecule that breaks apart creates additional gas-phase particles, increasing the total pressure.

In the power sweep, our Δp is ∼20% higher than the paper's. This overestimate traces directly to the power coupling efficiency η. We use $\eta = 0.50$, meaning 50% of the source power reaches the electrons. If the true coupling is closer to 0.40, the dissociation rate would decrease and Δp would drop by ∼20%, matching the paper. The paper assumes $P_{\text{abs}} = P_{\text{source}}$ (i.e. $\eta = 1$) but their unpublished C++ code may include an internal calibration factor. Ryan & Plumb (1990) noted that they needed to reduce their calculated dissociation rates by a factor of 2 to match experiments, attributing the discrepancy to matching network and sheath losses—the same physics that η captures.

In the pressure sweep, the model correctly reproduces the non-monotonic shape: Δp rises at low p_{OFF}, peaks near 2.5–3 Pa, and then declines. This turnover is the key experimental signature of SF_x deposition on the reactor walls (reactions S32–S40). Without deposition, Δp would increase monotonically. The peak in our model is shifted to slightly higher p_{OFF} (∼2.5 Pa vs ∼1.5–2 Pa in the paper), suggesting that our deposition rate may be slightly underestimated at intermediate pressures. The deposition probability of 0.03 is a fitted parameter in the paper; a value of 0.04–0.05 would shift our peak to lower p_{OFF}.

13 Neutral Species Composition

13.1 SF_6

SF_6 is the parent gas and its density is determined by the competition between the feed rate, pumping, and electron-impact dissociation. In the power sweep, our SF_6 density is higher than the paper's at high power (1.0×10^{20} vs $5 \times 10^{19}\,\text{m}^{-3}$ at 3500 W), meaning our model dissociates SF_6 less aggressively. This is a direct consequence of $\eta = 0.50$: with half the power reaching the electrons, the dissociation rate ($\propto n_e \times k_{\text{diss}}(T_e)$) is proportionally lower. In the pressure sweep the agreement is better because at fixed 2000 W, the variation is driven by the SF_6 loading rather than the power.

13.2 F

Fluorine atoms are the primary etchant species and are produced by every dissociation reaction (G1–G7). Our F density is ∼0.5–0.7× the paper's value across both sweeps. This follows directly from the lower SF_6 dissociation rate: fewer SF_6 molecules break apart, so fewer F atoms are produced. The near-linear increase of n_{F} with power is correctly reproduced because $n_{\text{F}} \propto n_e \propto P_{\text{abs}}$ in the regime where F loss (surface adsorption + recombination with SF_5) scales linearly with n_{F}.

13.3 F_2

F_2 is not produced by electron-impact dissociation of SF_6 in this model (the paper considered but rejected F_2 as a direct dissociation product). Instead, F_2 is produced almost entirely by surface recombination: F + F(s) → F_2 + bare (reaction S8). Since this rate depends on $n_{\text{F}} \times \theta_{\text{F}}$, and our n_{F} is ∼0.6× the paper's, the F_2 production rate is correspondingly lower. The ∼0.5× offset in F_2 is therefore a downstream consequence of the F offset, not an independent error.

13.4 SF_5

SF_5 shows the best agreement of all neutral species (0.5–1.0×) and its remarkably flat profile is reproduced in both sweeps. This flatness is a key physical prediction: SF_5 is the primary dissociation product of SF_6 (G1) but is immediately consumed by recombination with F (G35: SF_5 + F → SF_6). The Troe fall-off formula gives $k_{\text{G35}} \approx 9.1 \times 10^{-18}\,\text{m}^3/\text{s}$ at 0.921 Pa, which is in the high-pressure limit. This fast recombination acts as a chemical buffer: any increase in SF_5 production is immediately counteracted by faster consumption, keeping SF_5 nearly constant. The fact that our model reproduces this buffering confirms that the Troe formula from Ryan & Plumb (1990) is correctly implemented.

13.5 SF_4 and SF_3

SF_4 is produced by single-step dissociation (G2), multi-step dissociation via SF_5 (G4), and gas-phase recombination of SF_3 with F (G37). Its agreement (0.6–1.0×) is good. SF_3 is the least abundant SFx radical in both models, confirming the paper's statement that fragments smaller than SF_3 (i.e.

SF_2, SF, S) can be neglected. The G37 rate for $SF_3 + F \rightarrow SF_4$ is in the fall-off regime at 0.921 Pa ($k \approx 2.5 \times 10^{-18}\,m^3/s$), which is fast enough to keep SF_3 density low. The G36 rate for $SF_4 + F \rightarrow SF_5$ is in the low-pressure limit ($k \approx 6.4 \times 10^{-20}\,m^3/s$), which is 150× slower than G35. This enormous difference—caused by the different pressure regimes of the three Troe reactions—is why SF_4 accumulates to relatively high densities while SF_5 does not.

14 Charged Species

14.1 SF_5^+ and F^-

The model correctly identifies SF_5^+ as the dominant positive ion and F^- as the dominant negative ion across all conditions. Both of these are produced by the highest-cross-section electron-impact reactions on SF_6: dissociative ionization (G8, threshold 15.5 eV) and dissociative attachment (G17, threshold 0.1 eV) respectively. The agreement for F^- is excellent (0.8–1.2×), which is significant because F^- is the main negative charge carrier and directly affects the electronegativity.

14.2 Electron Density

Our n_e is ∼1.5–2× higher than the paper's in the power sweep. This is counterintuitive given that we use $\eta = 0.50$ (which should reduce n_e), but is explained by the h-factor correction. The full electronegative h-factors reduce the ion wall flux by a factor of ∼5 at $\alpha \approx 7$. This means ions (and hence electrons, by ambipolar coupling) are confined more effectively, increasing n_e for a given power input. The two competing effects (η reducing power input, h-factors reducing losses) partially cancel, leaving n_e within 2×.

In the pressure sweep, n_e agrees more closely because at higher pressure the ion confinement is already strong (high collisionality), and the h-factor correction is less significant.

14.3 Electronegativity ($\alpha = n^-/n_e$)

The electronegativity ratio α shows the largest discrepancy (2–3× in the power sweep). This is the most sensitive quantity in the model because it is the ratio of two quantities (n^- and n_e) that each depend on different loss mechanisms. A small change in the ion wall loss rate shifts n_e without affecting n^- (which is confined by the sheath and has no wall loss), amplifying the ratio.

The h-factor prefactor $(1 + 3\alpha/\gamma)/(1 + \alpha)$ was derived by Lee & Lieberman (1995) as a heuristic interpolation between the electropositive limit ($\alpha = 0$) and the strongly electronegative limit ($\alpha \gg 1$). In the intermediate regime (α = 2–10, which is exactly where our model operates at 2000 W), the interpolation is acknowledged by the authors to be approximate. More sophisticated treatments (e.g. the full spatial profiles of Lichtenberg et al. 1994, or the Thorsteinsson & Gudmundsson 2009 formulation) could improve the α prediction but were not implemented because the paper explicitly cites only the Lee & Lieberman formulation.

Despite the absolute offset, the qualitative behavior of α is correct: hyperbolic decrease with power, nearly linear increase with pressure, and the magnitudes are in the right ballpark (∼5–60 in our model vs ∼1–45 in the paper).

15 Coupled Nature of the Discrepancies

The discrepancies are not independent—they are coupled through the power balance and trace primarily to a single root cause: the effective power delivered to the electrons.

The causal chain is:

1. $\eta = 0.50$ reduces the power available for electron heating.
2. Lower effective power → lower n_e (fewer electrons sustained).
3. Lower n_e → lower dissociation rate → higher SF_6 density, lower F density.
4. Lower F → lower F_2 (less surface recombination feedstock).
5. However, the electronegative h-factors partially compensate: reduced ion wall loss → higher n_e than η alone would predict.

6. The partial compensation leaves n_e within 2× but pushes α higher (because n^- is unaffected while n_e is modified).

This coupling means that improving η (e.g. from 0.50 to 0.65) would simultaneously improve SF_6, F, F_2, and Δp by bringing the dissociation rate closer to the paper's value. It would not, however, fix α, because that depends on the h-factor formulation rather than the power balance.

16 What Would Be Needed for Exact Reproduction

Exact quantitative reproduction of the paper's figures would require information that is not available in the published paper:

1. **The exact value of η.** The paper does not state whether any power correction is applied.
2. **The C++ source code.** The paper uses a Newton–Raphson solver, which may find a different root than our ODE time-integration approach if the system has multiple steady states.
3. **The exact h-factor implementation.** The paper's companion C_4F_8 paper (Ref. [36]) uses the Lee & Lieberman formulation but applies it to a two-chamber reactor with separate h-factors for each chamber. The details of this two-chamber treatment are not fully specified.
4. **Numerical tolerances and convergence criteria.** The Newton–Raphson method is sensitive to initial guesses and convergence thresholds, which can affect which steady state is found.

None of these factors affect the *physics* of the model—they are implementation details. The fact that our independent Python implementation, using a completely different solver method (ODE time-integration vs. algebraic root-finding), reproduces all qualitative trends and achieves quantitative agreement within the model's sensitivity band is strong evidence that the underlying physics is correctly captured.

17 Conclusion

The Python implementation of the Kokkoris SF_6 global model correctly reproduces all 13 qualitative trends from the published paper across both power and pressure sweeps. Ten of twelve benchmarked quantities agree within the model's inherent 2× sensitivity band. The two quantities rated "Fair" (electronegativity α) trace to a known approximation in the Lee & Lieberman h-factor formulation in the electronegative transition regime, not to errors in the chemistry, the Troe fall-off rates, or the solver.

The remaining quantitative offsets are coupled and trace primarily to the power coupling efficiency η, which sets the overall scale of the electron density and dissociation rate. This is a single adjustable parameter that could be tuned to match specific experimental systems, but was deliberately set to a physically reasonable value ($\eta = 0.50$) rather than fitted to the paper's curves.

The model is suitable for use as a validated computational tool for SF_6 plasma research, including extension to SF_6/O_2 mixtures, BOSCH process simulation, and reactor design optimization.

18 References

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